

SITANSHU SINGH

Data Analyst | Computer Science and Data Analytics, IIT Patna | SPI: 9.00 / 10
+91 8115926935 | official.sitanshu369@gmail.com | LinkedIn | GitHub | Portfolio

Education

Indian Institute of Technology Patna

2023 to 2027

Bachelor's in Computer Science and Data Analytics | SPI: 9.00 / 10

Coursework: Data Structures and Algorithms, DBMS, Operating Systems, OOP, Machine Learning, Probability and Statistics, Linear Algebra, Discrete Mathematics

Technical Skills

Programming: Java, Python (Pandas, NumPy, Matplotlib, Seaborn), SQL

SQL: Joins, CTEs, Subqueries, Window Functions, Aggregations, Data Cleaning

Data Analytics: Data Cleaning, Data Wrangling, Exploratory Data Analysis, Statistical Analysis, Data Processing, Data Quality Validation, Geospatial Data Validation

BI and Visualization: Power BI, Advanced Excel, Dashboard Development, Data Visualization, Reporting, KPI Tracking

Specialized Tools: Roboflow, COCO JSON, Google Earth Engine

Databases: MySQL

Generative AI: LLMs, Prompt Engineering, RAG, LangChain, OpenAI API

Version Control: Git, GitHub

Experience

Data Analytics Research | NIT Srinagar, Srinagar, India (On-site)

Jun 2026 to Aug 2026

- Prepared and validated 2,500+ field images and 14,300+ annotated waste instances for the Dal Lake Plastic Waste Detection project, using Python for data cleaning ahead of computer vision model development.
- Ran exploratory data analysis on annotation data in Python (Pandas, Matplotlib, Seaborn) to identify class distribution, object density, and consistency issues across 6 plastic waste categories.
- Designed Power BI dashboards with 15+ KPIs to track annotation progress and dataset composition, and automated analysis of COCO JSON files to support dataset validation.

Data Analyst | Buddiz AI, UAE (Remote)

May 2026 to Jul 2026

- Investigated environmental datasets from Sentinel-2, MODIS, and ground observations for the UAE Carbon Sequestration Mapping project, estimating carbon storage across UAE mangrove ecosystems.
- Processed, cleaned, and standardized 50,000+ environmental data records in Python (Pandas, NumPy), resolving missing values to prepare analytical layers for downstream analysis.
- Created Power BI dashboards with 20+ KPIs to monitor carbon estimation metrics and vegetation health, producing 15+ supporting visualizations in Matplotlib and Seaborn.

Data Analytics Research | IIT Bhilai, Bhilai, India (On-site)

Jan 2026 to Apr 2026

- Cleaned 100,000+ sensor records and annotated 2,500+ industrial images for YOLOv8 model development, using Python (Pandas, NumPy) to prepare high quality datasets for predictive maintenance in steel manufacturing operations.
- Conducted EDA to identify the most fluctuating sensors, using Matplotlib and Seaborn to surface trends and anomalies across 15+ visualizations.
- Developed Power BI dashboards with 15+ KPIs to monitor sensor behavior and equipment performance, and automated data validation workflows to cut manual preprocessing effort.

Projects

Delivery Performance Analysis for a Multi Zone Food Platform | Python, Excel, Power BI

- Analyzed close to 50,000 simulated delivery records with Python and Power BI, building drill downs by zone, time of day, and rider that showed two zones accounted for a disproportionate share of late evening deliveries, and recommended shifting rider allocation to those windows.

Customer and Sales Analysis for an Online Retailer | SQL, Python, Power BI, Excel

- Used SQL joins and window functions across roughly 20,000 transaction records to build customer summaries, then segmented customers in Python by spend and frequency.
- Built a Power BI and Excel view for the category and pricing team, highlighting which customer segments were most sensitive to discounting.

Certifications

- Microsoft Excel (Microsoft, Coursera): covered Excel formulas, PivotTables, XLOOKUP, and data cleaning with Copilot
- Database Structures and Management with MySQL (Meta, Coursera): covered relational database design, primary and foreign keys, and MySQL stored procedures
- Data Analysis with Python (IBM, Coursera): covered data wrangling, exploratory data analysis, and regression modeling with Pandas, NumPy, and Scikit-learn

Achievements

Solved 274+ problems on LeetCode (102 Easy, 126 Medium, 46 Hard) with a 60 day max streak.

Received a Letter of Recommendation from Dr. Veningston K, Assistant Professor, Department of CSE, NIT Srinagar, for his work on the Dal Lake Plastic Waste Detection internship (2026).

Leadership and Activities

Media and PR Coordinator, TEDx IIT Patna: Wrote and edited outreach content and coordinated with design and logistics teams for event day.

Coordinator, National Service Scheme: Planned logistics and turnout for NSS IIT Patna campaigns of about 30 volunteers.